

Qualitative Questions

3. Would you use PingPong more if it were different? If not, why not? If yes, what would need to change about PingPong for you to use it more?

Category	Count
PingPong is counter productive (i.e. makes student more confused,	1
Wished PingPong functioned differently (find online resources for the student	1
Student would rather just stay away from LLMs generally in their education	1
Student did not care/dedicate time to use it	1

4. (For Reference) If you wanted to learn more about a statistical concept, would you prefer to use PingPong or another LLM?

5. Please elaborate on your response to the previous question. If you prefer one LLM over another, why? If you are indifferent, why?

Category	Count
PingPong, because it makes students explain concepts before helping them with the question. This helps the student figure out the question more on their own. On the other hand, other LLMs will just give the answer.	10
Prefer to use perplexity	1
I like PingPong, but it is more time consuming to get your answer compared to other LLMs	4
Another LLM, like ChatGPT, because it gives more direct answers with more information. Other LLMs give complete answers for the student to reference.	6
PingPong because I like how it is trained for the course specifically	11
I prefer PingPong because it gives justification for its answer and is more consistent compared to another LLM	1

I would prefer Pensieve	1
Other LLM, because PingPong isn't helpful to the student due to UI issues and the persistent check of understanding.	1
Indifferent, PingPong and other LLMs have separate use cases	3
Other LLM, because it is familiar to the student	1

7. What improvements or changes to PingPong would you like to see implemented? For example, do you find any features of PingPong frustrating or confusing? If yes, how would you improve those features?

Category	Count
Guiding Questions are excessive, sometimes it is a simple syntax or definition question that does not need a check of understanding	15
No changes	2
Larger text box to read PingPong's responses	1
PingPong gets lost in circles, cannot stay on topic to successfully answer the question	4
It would incorrectly say that the student was asking a question to an entire homework question (asking about approach for example)	2
Wants an audio component to give input	1
A way to make PingPong explain things at a lower level (less formal, simple examples, simple vocabulary)	1
Wishes PingPong generated an example question to test student knowledge	1
Wishes it was widely available across UC Berkeley	1
A bit slow	2
PingPong would keep earlier chats in mind	1

8. Do you think PingPong has any notable strengths relative to other LLMs like ChatGPT, Gemini, or Claude? If yes, which features do you think are the most impactful?

Category	Count
Asking for understanding, going step by step, helps the student actually learn compared to other LLMs	20
Friendly	1
Good at Explaining	2
Gives reliable information	3
No it does not	3
How it is based on course material	3
How it is provided and approved by the university	2
Not sure, I do not use other LLMs	1

10. Without LLMs, do you think your understanding of the STAT 131A material would be different? If yes, why do you think your understanding would be different? If not, why do you think your understanding would be the same?

Category	Count
No	5
No, I rarely/don't use LLMs	4
Yes, it helps a lot with the programming/lab assignments	3
Yes, it helps a lot with understanding conceptual aspects of the course	13
No, but I would have to spend more time on the course (i.e. going to OH more)	6
No, I use other materials to understand content while using PingPong for other use cases	5
Yes, I do not feel intimidated like if I went to OH and asked "dumb" questions	5

Yes, PingPong is so convenient, I can ask it questions whenever	2
Yes, I use it as a “crutch” instead of consulting course material	2

11. Without LLMs, do you think your homework and exam grades in STAT 131A would be different? If yes, why do you think your grades would be different? If not, why do you think your grades would be the same?

Category	Count
No	5
No, because tests were conceptual, but it helped slightly with homework	5
Yes	6
No, because I rarely use or dont use LLMs/Ping Pong	4
Not sure	3
Yes, I use PingPong/other LLM a lot to understand course material (many mentioned the convenience of PingPong to going to OH/asking on Ed)	20
No, because I would go to OH to fill the gaps if so / spend more time learning material	2

12. (For Reference)Which of the following would you prefer?

1. Programming+statistics courses allow students to use any LLM for homework help.
2. Programming+statistics courses only allow customized chatbots, similar to PingPong.
3. Programming+statistics courses do not allow students to use any LLM.

13. Please elaborate on your response to the previous question. Why do you prefer your chosen option?

Category	Count
Customized Chatbots, because they prompt the user for understanding and act as a tutor. Other LLMs would just give the answer	15

straightaway.	
All LLMs, customized chatbots and other LLMs have separate use cases that are beneficial. As long as the student is academically honest.	11
Customized Chatbots, they even the playing field for students (no students have extra advantages with advanced LLM usage)	1
Customized Chatbots, it is crucial to have it be adjusted to the course's material, and have it approved by the professor.	5
Customized Chatbots, even if they are not used, it is great to have it as another resource (One mentioned that it is more accessible when students have difficulties making it to campus)	2
All LLMs, people will use ChatGPT regardless. (Many mention that the real world has no restrictions on LLM usage, so students to learn to navigate it)	5
All LLMs, PingPong has errors that make it not sufficient (accusing student of cheating when referencing homework questions)	1
All LLMs, PingPong and other LLMs function similarly enough	1

14. What concerns, if any, do you have about AI in education? Has PingPong changed your perspective at all? For example, are you scared that LLMs may replace human tutors,

interpersonal interaction, or office hours? On the other hand, are you optimistic about the future of AI in education?

Category	Count
Students will use LLMs by not actually learning the material and using it to complete the work for them	15
I think AI serves other purposes in education better. (i.e. summarizing large topics)	1
AI is making us lose our creativity and original thoughts	2
Worried that LLMs will be used in place of human tutors. Human interaction is irreplaceable (One suggests the need for LLMs illustrate the need for more tutors)	5
Optimistic, LLMs lower the barrier of getting help in class (intimidation from asking tutors "dumb" questions)	1
Optimistic, but students need to be taught how to correctly use LLMs	3
Optimistic, sees how LLMs, like Ping Pong, aid in understanding the material	2
No	2